

Customer Targeting & Incremental Revenue Optimization

Deciding *who* to target with a marketing email to maximize incremental revenue — not just predicting who buys. Demonstration on the public, randomized Hillstrom dataset.

Targeting policy Score a customer [The story](#)

What this project shows

The question. A marketing email lifts conversions *on average* — but who should we actually send it to? The average can hide customers the email doesn't move, or even *annoys*. Targeting the right people is where the money is.

The method.

1. **Designed & validated the experiment** — power analysis, then balance checks confirming randomization held (treatment and control are statistical twins).
2. **Measured the average effect** — email lifts conversion 0.57% → 1.07% (~87% relative, $p \approx 4e-10$). Real, but *not constant* across customers (~3× spread by recency) — so an average is not a targeting rule.
3. **Modeled the individual effect** — S-, T-, and a hand-rolled **X-learner**.
4. **An honest pivot.** On the rare *conversion* outcome, no model beat random (Qini ≈ 0) — the signal was too sparse. I diagnosed it as outcome density (not algorithm), and switched to the denser *visit* outcome, where models beat random and the **X-learner won** (consistent with its robustness to the 2:1 class imbalance).
5. **Turned scores into a decision** — ranked customers by uplift, found the profit-maximizing cutoff, and showed targeting beats same-size random sending by 2×.

The judgment. The most important result is #4: knowing when *not* to trust a model is part of the analysis. Reported honestly, a model that doesn't beat random is a finding — and it led to the principled fix.

Built end-to-end: experiment design → EDA/balance → ATE → uplift models → Qini evaluation → targeting policy. Code at the [project repo](#).